ML Tutorial-2

A mobile service provider wants to classify SMS messages as Spam or Not Spam using a simple neural network (Perceptron)

X_1 = specific words & phrases

X_2 = Links & Attachments

X_3 = Bias

X_1	X_2	X_3	Target (y)
4	2	1	1
1	0	1	0
6	3	1	1
2	1	1	0

1) Inputs: X_1 & X_2

2) weights w_1 w_2

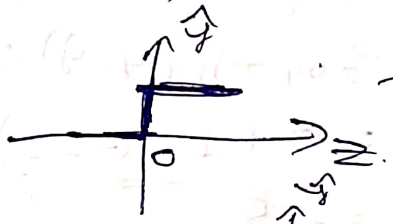
3) bias (b)

4) weighted sum: $z = x_1 w_1 + x_2 w_2 + b$

$$= \sum_{i=1}^{2} x_i w_i + b$$

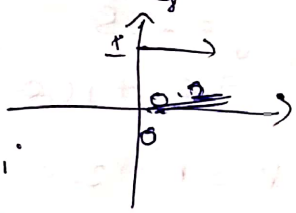
5) Activation function
(Here step function)

$$\hat{y} = \begin{cases} 0, & z < 0 \\ 1, & z \geq 0 \end{cases}$$



Sigmoid function.

$$\sigma(z) = \frac{1}{1+e^{-z}}$$



6) Updated weights

$$w_i^{new} = w_i^{old} + \eta (y - \hat{y}) x_i$$

η = Learning Rate = 1

y = Actual or True Target

$\hat{y}$ = Predicted Target

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i) $X_1 = 4$, $X_2 = 2$, $b = 1$

Let $w_1 = 0$, $w_2 = 0$

$z = X_1 w_1 + X_2 w_2 + b$

$= 4(0) + 2(0) + 1$

$= 1$

$\hat{y} = 1$ ✓

Iteration = Epoch => 10.

X_1	X_2	X_3	Target (y)
5	3	1	0
1	0	1	0
7	5	1	1
2	1	1	0
6	4	1	1

i) $X_1 = 5$ $X_2 = 3$ $b = 1$

Let $w_1 = 0$, $w_2 = 0$

$z = X_1 w_1 + X_2 w_2 + b$

$= 0 + 0 + 1$

$z = 1$

$\hat{y} = 1$ X but $y = 0$.

$w_1^{new} = w_1 + \eta (y - \hat{y}) X_1$

$= 0 + 1(0 - 1)5$

$= -.5$

$w_2^{new} = 0 + 1(0-1)3 = -3$

$\Rightarrow$ $X_1 = 1$, $X_2 = 0$, $b = 1$

$w_1 = -5$ $w_2 = -3$

$z = -.4$ $\hat{y} = 0$ ✓

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Time on site (X_1)	Items viewed (X_2)	Bias (x_3)	Target Purchase = 1 No = 0 (y)
2	3	1	0
15	25	1	1
8	12	1	1
3	5	1	0
12	20	1	1

i) $X_1 = 2$ $X_2 = 3$ $\cancel{x_3 = 1}$ $b = 1$

Let $w_1 = 0$ $w_2 = 0$ $\cancel{w_3 = 0}$

$z = X_1 w_1 + X_2 w_2$

$= 0 + 0 + 1 = 1$

$\hat{y} = 1$ ✗

$w_1^{new} = w_1^{old} + \eta(y - \hat{y}) x_1$

$= 0 + 1(0 - 1)2$

$= -2$

$w_2^{new} = 0 + 1(0 - 1)3 = -3$

ii). $x_1 = 15$ $x_2 = 25$ $b = 1$

$w_1 = -2$ $w_2 = -3$

$z = x_1 w_1 + x_2 w_2 + b = -30 - 75 + 1 = -104$

$\hat{y} = 0$ ✗

iii). $X_1 = 8$ $X_2 = 12$ $b = 1$

$\cancel{w_1 =}$

$\cancel{z}$

$w_1^{new} = -2 + 1(1 - 0)15$

$= -2 + 15 = 13$

$w_2^{new} = -3 + 1(1 - 0)25 = 22$

iii) $X_1 = 8$ $X_2 = 12$ $b = 1$

$w_1 = 13$ $w_2 = 22$

$z = 104 + 264 + 1 = 369$

$z > 0$ $\hat{y} = 0$ ✓

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